Mussa Phiri

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Professional Summary

Computer Science MSc graduate with Distinction and over three years of university teaching and tutoring experience in programming, distributed systems and cybersecurity. Research experience in AI/ML for cybersecurity, including MSc work on concept drift in audio-based Android malware classification and detection.

Education

MSc in Computer Science *Feb 2024 – Dec 2025*
University of Zululand, South Africa Distinction (78%)

- **Research Topic:** *Exploring the Impact of Concept Drift on Audio-Based Features in Android Malware Detection.*

BSc Honours in Computer Science *Feb 2023 – Dec 2023*
University of Zululand, South Africa

- **Research project:** *Metaheuristic-Optimised Firmware Malware Detection Using Audio-Based Features (Distinction 78%).*

BSc in Computer Science and Physics *Feb 2020 – Dec 2022*
University of Zululand, South Africa

Employment History

Part Time Lecturer *Current*
Eduvos, South Africa

- Lecture C#, Advanced Java (networking) and PHP, including lesson preparation, assessment design and marking, student consultation, and at-risk monitoring.

Cybersecurity Lab Instructor (Webgoat) *2025*
University of Zululand, South Africa

- Tutored students in hands-on cybersecurity labs using WebGoat, covering common web application vulnerabilities, including SQL injection, XSS, and broken access control.
- Guided students in using Burp Suite, SQL, and Python within Linux-based lab environments to analyse and demonstrate web application vulnerabilities.

Distributed Systems Lab Instructor *2024;2025*
University of Zululand, South Africa

- Led practical sessions on distributed systems in which students implemented client–server communication, concurrency, and multithreading in Python, using Docker to simulate the distributed environment.

Academic Student Assistant – Python & Java *2023;2024;2025*
University of Zululand, South Africa

- Supported Python and Java programming tutorials for more than 200 first-year students.
- Helped students with programming exercises, debugging, and assessments.

Writing Centre Assistant *2025*
University of Zululand, South Africa

- Supported students with academic writing, research proposals, reports, and referencing.

Skills

- **Languages:** Python, SQL, Java, C#
- **Machine Learning:** classification, regression, forecasting, anomaly detection, model evaluation, temporal analysis
- **Data Analytics & Visualisation:** data cleaning, exploratory data analysis (EDA), KPI development, Power BI, Tableau
- **Software Engineering:** SDLC, CI/CD, unit & integration testing, Git/GitHub, API & integration development (MuleSoft)
- **Tech Stack:** Azure, Jupyter Notebook, Google Colab, MS Excel

Publications

Conference Proceedings

- **M. Phiri**,P. Tarwireyi and A. Terzoli. 2026. "Temporal Adaptation through Sliding Window Retraining for Audio-Based Obfuscated Android Malware Classification" In *Proceedings of the Southern Africa Telecommunication Networks and Applications Conference (SATNAC)*, Drakensberg, South Africa.

- **M. Phiri**, P. Tarwireyi and A. Terzoli. 2025. "Investigating the Robustness of Audio Features Under Concept Drift in Android Malware Detection." In *Proceedings of the Southern Africa Telecommunication Networks and Applications Conference (SATNAC)*, Hermanus, South Africa.
  - **Award:** 3rd Best Paper Award, SATNAC 2025.

- L. Mkhize, P. Tarwireyi, **M. Phiri**, and M. Adigun. 2024. "Comparison of MFCC, BFCC, and NGCC Audio-Based Features in Android Malware Detection." In *Proceedings of the Southern Africa Telecommunication Networks and Applications Conference (SATNAC)*, Skukuza, South Africa.

Under Review

- **M. Phiri** et al. "Android Malware Detection under Concept Drift: A Systematic Review." *Springer: Journal of Computer Virology and Hacking Techniques*.

Research Grants and Relevant Awards

- **1st Place** - Hack4Dev UKZN Data Science Hackathon, South Africa, 2023.
- **Telkom Departmental Bursary** - University of Zululand, Telkom Centre of Excellence, South Africa, 2024.

Certifications

- Google Cybersecurity Professional Certificate
- MuleSoft 4.X Complete Guide For Beginners – Hands-On Projects
- AWS Academy Cloud Foundations

References

- **Dr Paul Tarwireyi**
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  University of Zululand, South Africa
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- **Prof Alfredo Terzoli**
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